

Manpreet Singh

AI Engineer

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in [manpreet-singh-cse](#)

📍 Canada

Summary

AI/ML Engineer with 5+ years of experience delivering deep learning and NLP solutions in production environments. Proven ability to drive business outcomes, including a 25% productivity improvement through AI automation. Bridging industry and research, with recent focus on LLM interpretability and efficient AI.

Education

Master of Computer Science

Jan 2023 – Dec 2024

Dalhousie University – Halifax, NS, Canada

Thesis – Interpreting the effect of Quantization on LLMs

Relevant coursework – Machine Learning, Deep Learning, Natural Language Processing and ML for Software Engineering Applications.

GPA (Graduate courses) – 3.9/4.3

Professional Experience

Research Assistant

May 2023 – Dec 2024

Hypermatrix Labs, Dalhousie University – Halifax, NS, Canada

Interpreting the effects of Quantization on LLMs

- Conducted first of kind research by investigating multiple LLMs across 3 quantization settings, analyzing its impact on model confidence and calibration.
- Focused further research on neuron behavior by analyzing redundancy, dead neurons and overall contribution.
- Utilized **Integrated Gradients** to identify and analyze **10000+** neurons driving model predictions, measuring the effect of model sensitivity due to features required for model prediction.
- Developed LLM-based technique to generate human-readable explanations for neuron activations, reducing explanation validation cost by 10%.
- Proved utilization of quantization techniques to be reliable in high-risk and resource-constraint environments.
- Published findings in a thesis, contributing to the growing body of research on LLM transparency and explainability.

Manifold Learning in NLP Embeddings

- Evaluated **5+ manifold learning techniques** for dimensionality reduction of NLP token embeddings.
- Visualized and analyzed embedding spaces to identify model-learned biases.
- Delivered actionable insights on algorithm performance, enabling more equitable and unbiased NLP model training pipelines.

AI Engineer

Jul 2019 – Oct 2022

iGnite Labs, Tata Consultancy Services – Chennai, India

- Developed Natural Language to SQL converter using GPT-3, processed 500+ daily queries with 92% accuracy.
- Deployed Rasa-powered NLP chatbot handling 1000+ daily developer queries, improving productivity by 25%.
- Created Computer Vision pipeline converting 2D blueprints to 3D models, reducing visualization time by 20%.
- Mentored 100+ engineers in ML tools and best practices; 95% training completion rate.
- Automated 15+ dev tasks using Python and shell scripting, saved 20+ hours weekly.

Key Skills

- **Programming Languages:** Python (primary), JavaScript, Java, C.
- **Machine Learning & AI:** PyTorch, TensorFlow, Model training & finetuning, Transformers, LLMs, RAG, Prompt Engineering, NLP/NLU, Rasa, Explainable AI, CNN, GANs, Scikit-learn, Model Interpretability.
- **Data Analysis:** Pandas, NumPy, SQL, EDA, Statistics, Data Visualization.
- **High-Performance & Distributed Computing:** Slurm, Multicore Optimization, Cluster Management.
- **Software Development & DevOps:** Docker, Git, Flask, Django, RESTful APIs.
- **Management & Leadership:** Agile Methodologies, Technical Project Management, Team Leadership, Mentorship.

Certifications

- [Getting Started with Deep Learning - Nvidia](#) | [LLMOps](#)

Extracurricular Achievements

- Presented Neural Style Transfer project to India's Education Minister (TCS internal showcase).
- First place, AI-themed Quiz – L.K. College, Jalandhar.
- Completed 20-week intensive AI/ML training (Top 0.5% among 1000 participants).

References

Available upon request.